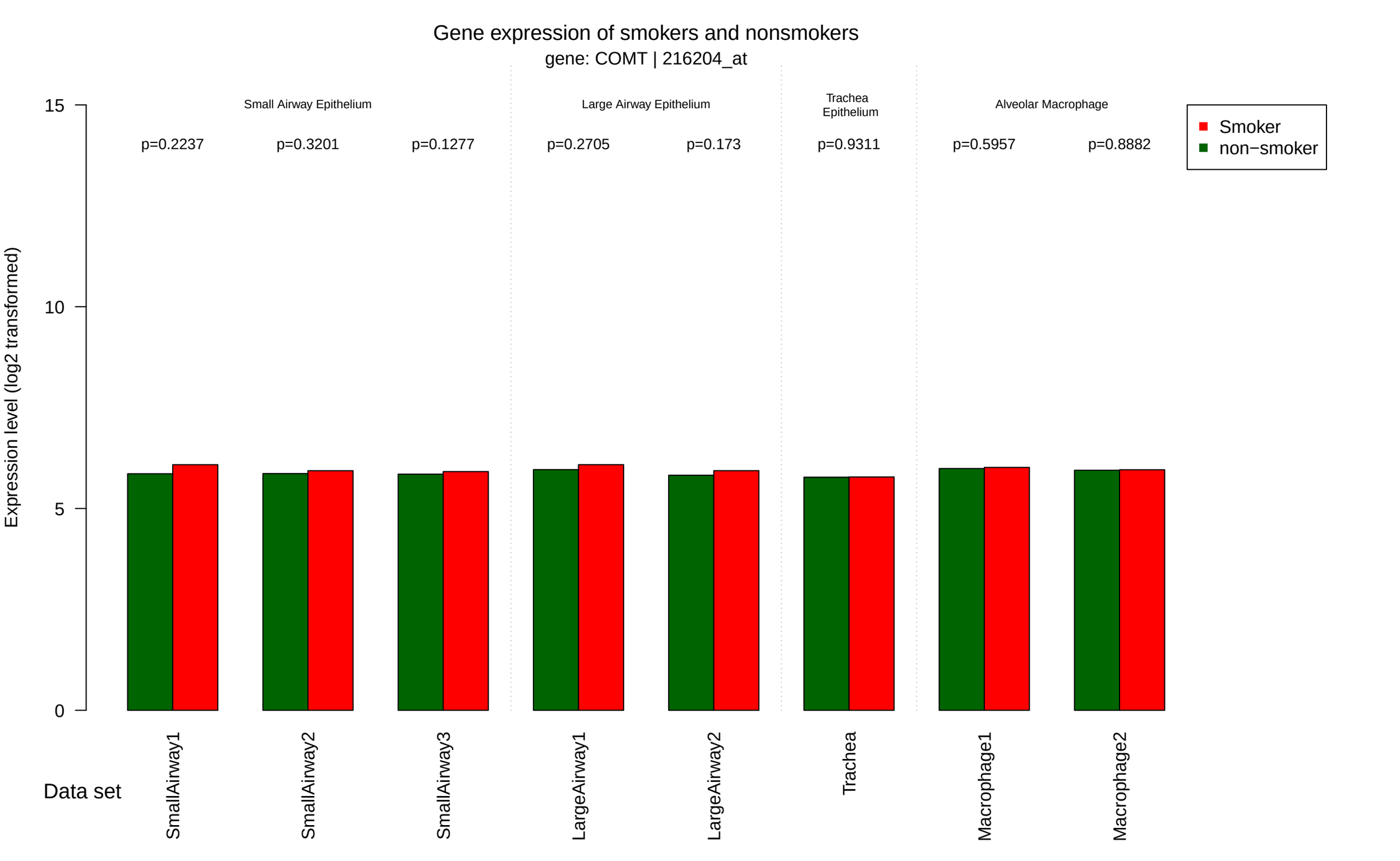
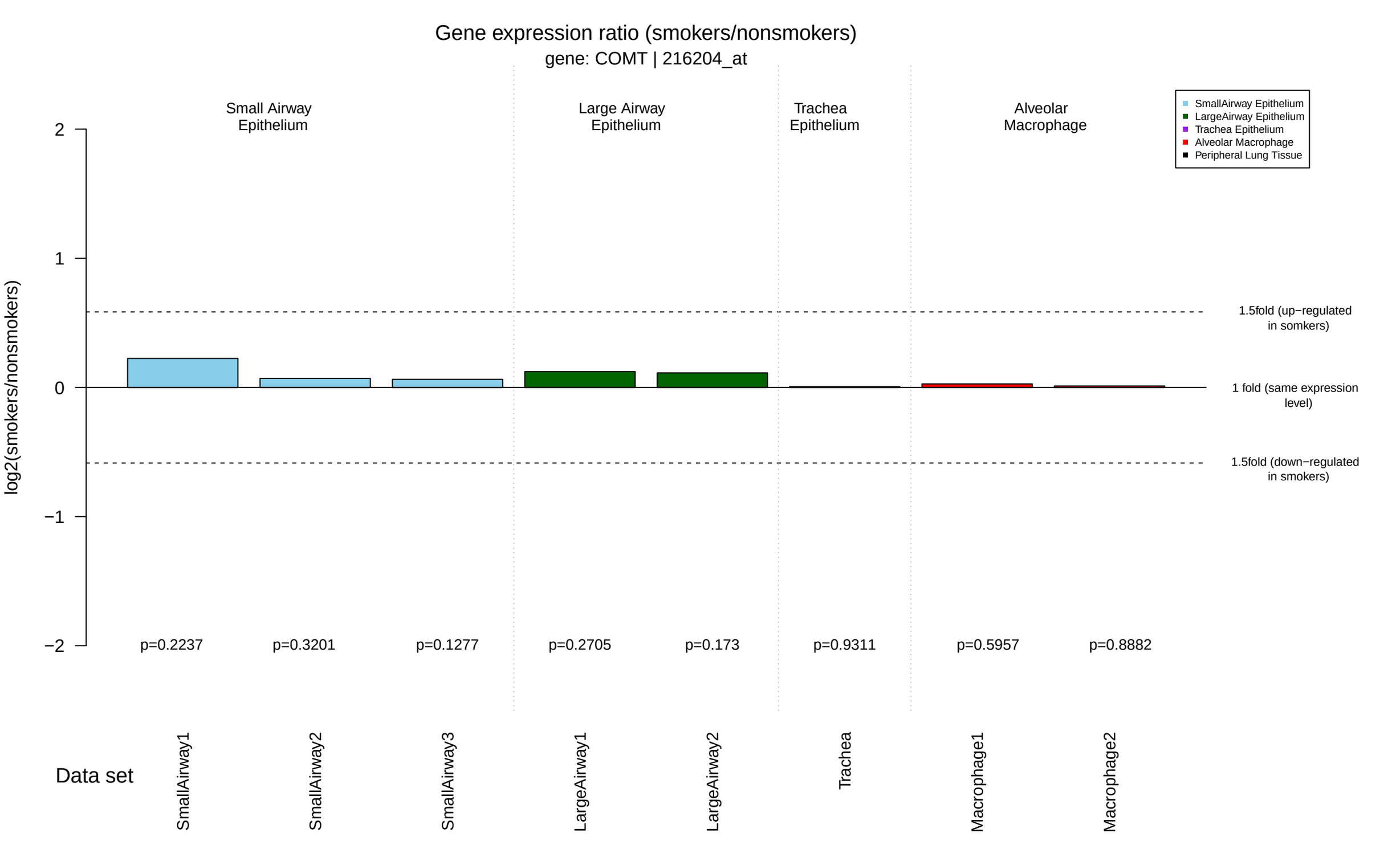
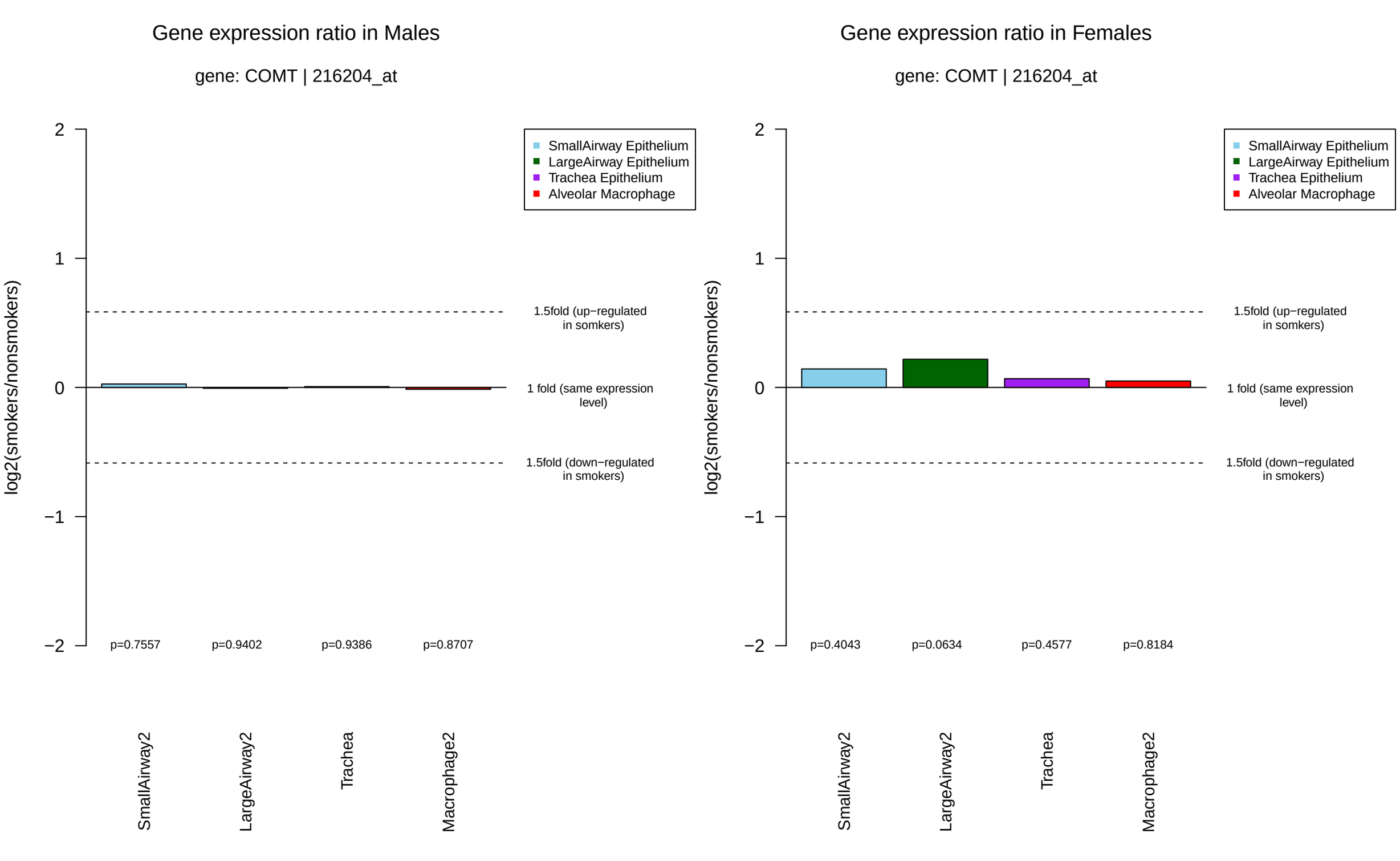
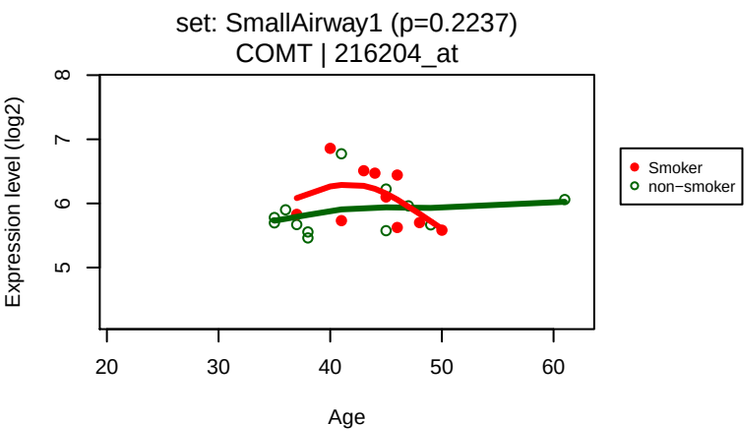
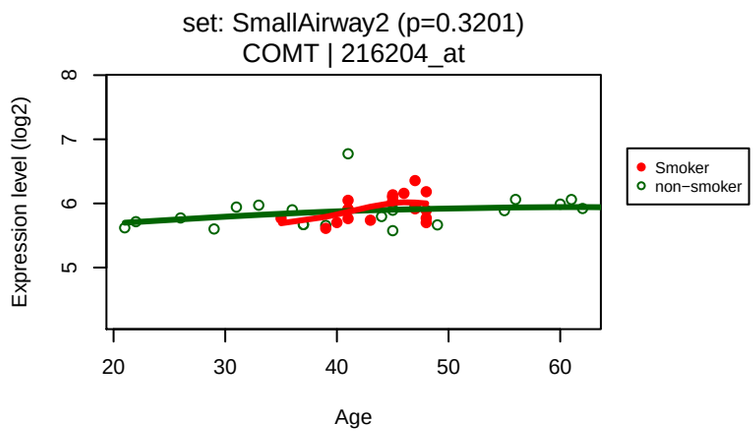
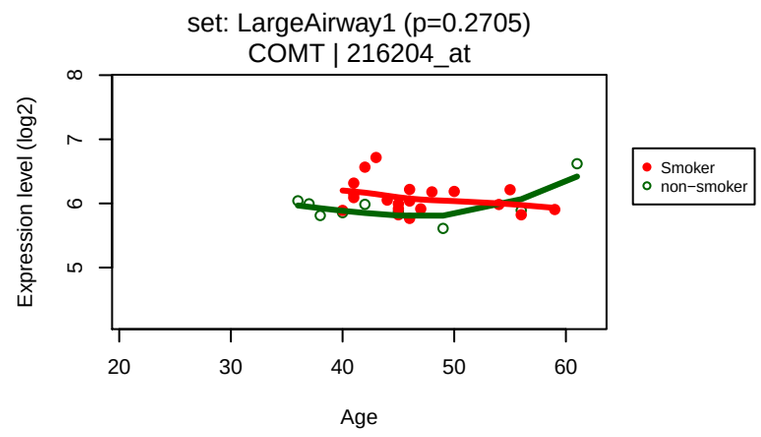

Gene expression of smokers and nonsmokers
gene: COMT | 216204_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2237
p=0.3201
p=0.1277
p=0.2705
p=0.173
p=0.9311
p=0.5957
p=0.8882
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

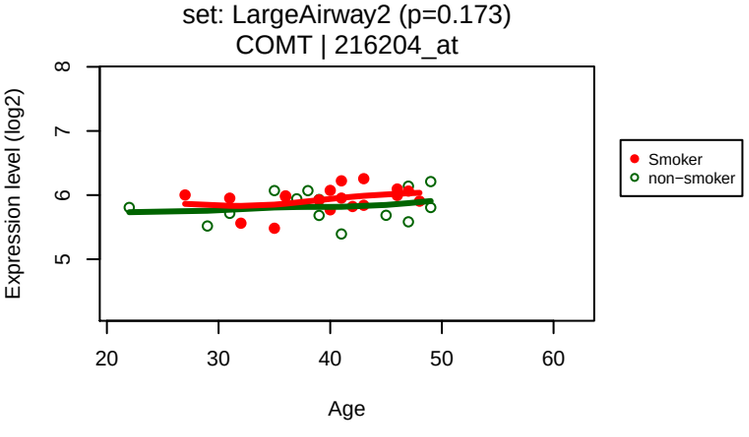

Gene expression ratio (smokers/nonsmokers)
gene: COMT | 216204_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2237
p=0.3201
p=0.1277
p=0.2705
p=0.173
p=0.9311
p=0.5957
p=0.8882
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

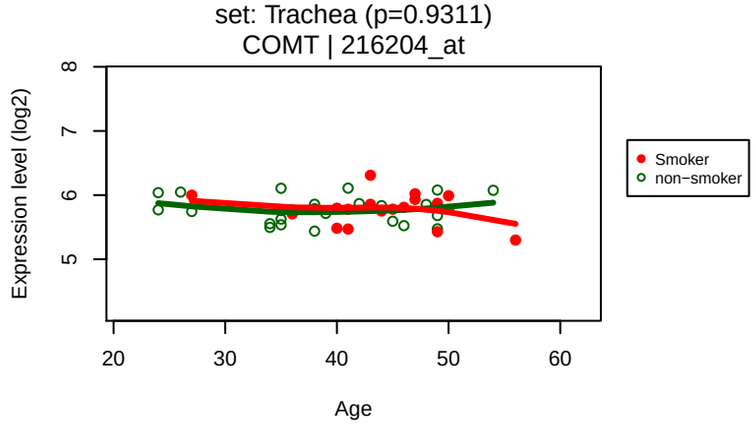

Gene expression ratio in Males
gene: COMT | 216204_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7557
p=0.9402
p=0.9386
p=0.8707
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: COMT | 216204_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4043
p=0.0634
p=0.4577
p=0.8184
SmallAirway2
LargeAirway2
Trachea
Macrophage2

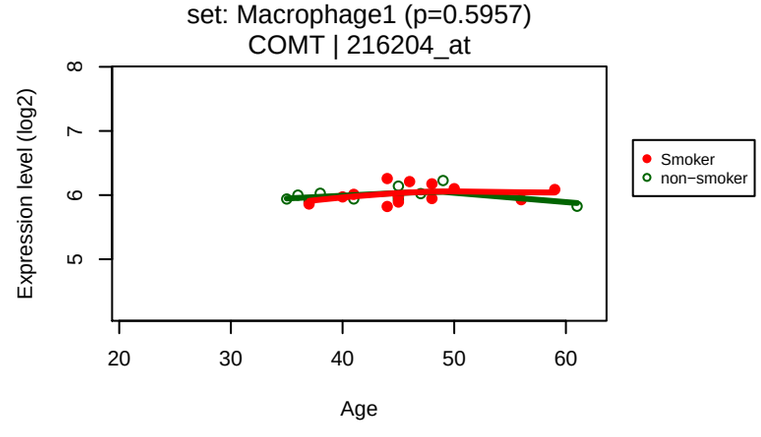

set: SmallAirway1 (p=0.2237)
COMT | 216204_at
Expression level (log2)
Age
Smoker
non−smoker

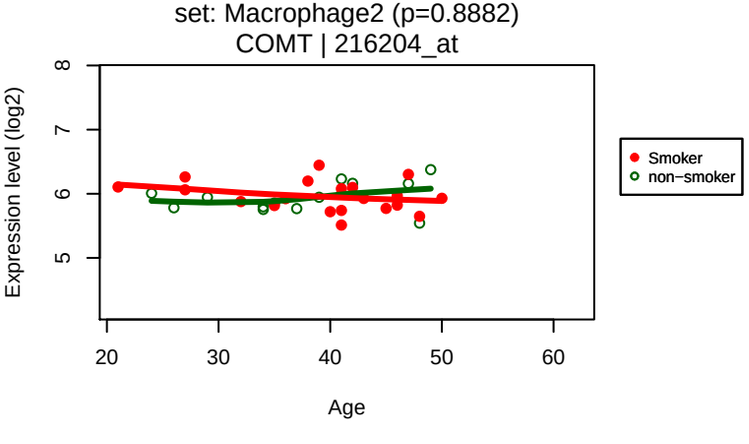

set: SmallAirway2 (p=0.3201)
COMT | 216204_at
Expression level (log2)
Age
Smoker
non−smoker

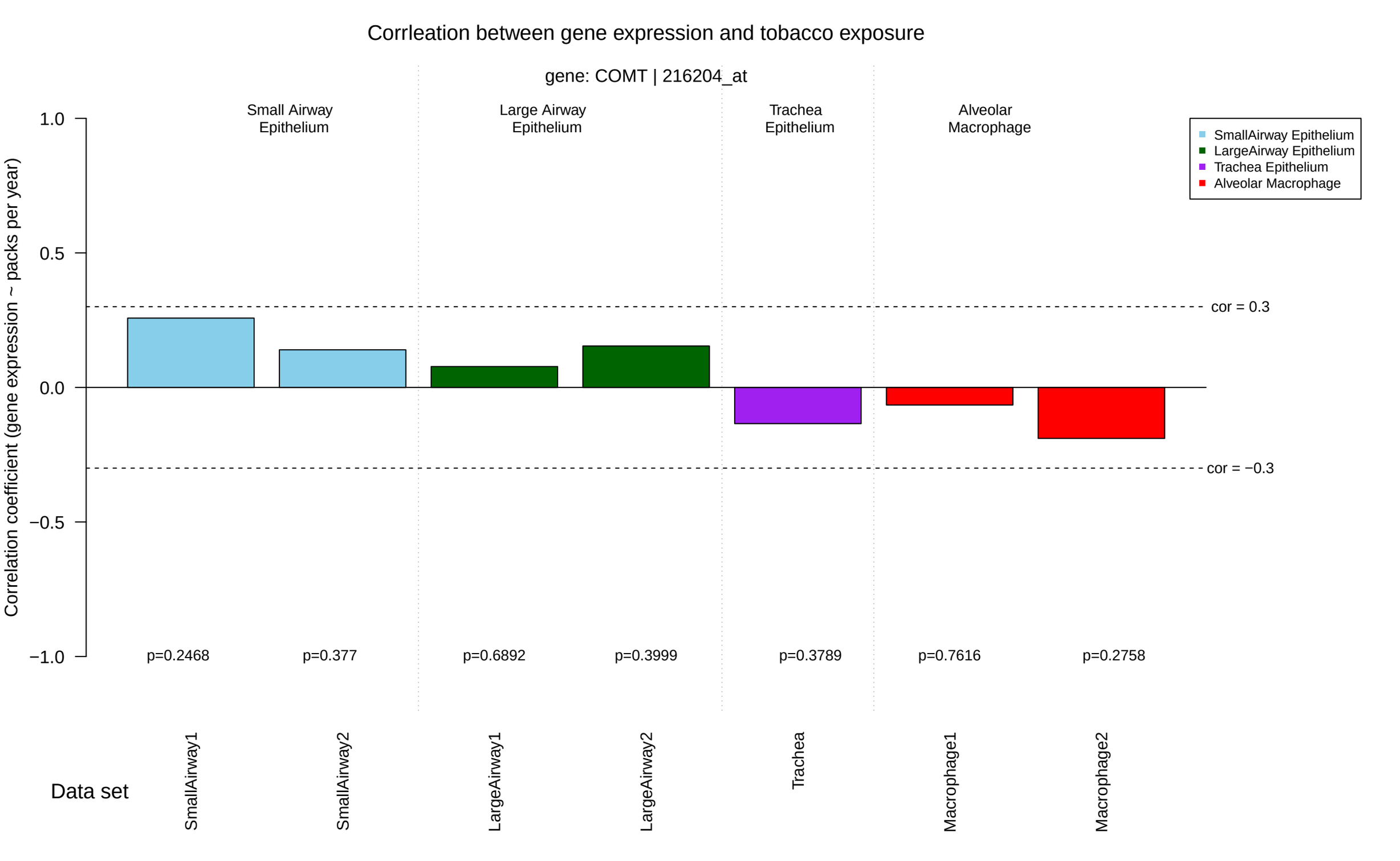

set: LargeAirway1 (p=0.2705)
COMT | 216204_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.173)
COMT | 216204_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9311)
COMT | 216204_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5957)
COMT | 216204_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8882)
COMT | 216204_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: COMT | 216204_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2468
p=0.377
p=0.6892
p=0.3999
p=0.3789
p=0.7616
p=0.2758
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2